

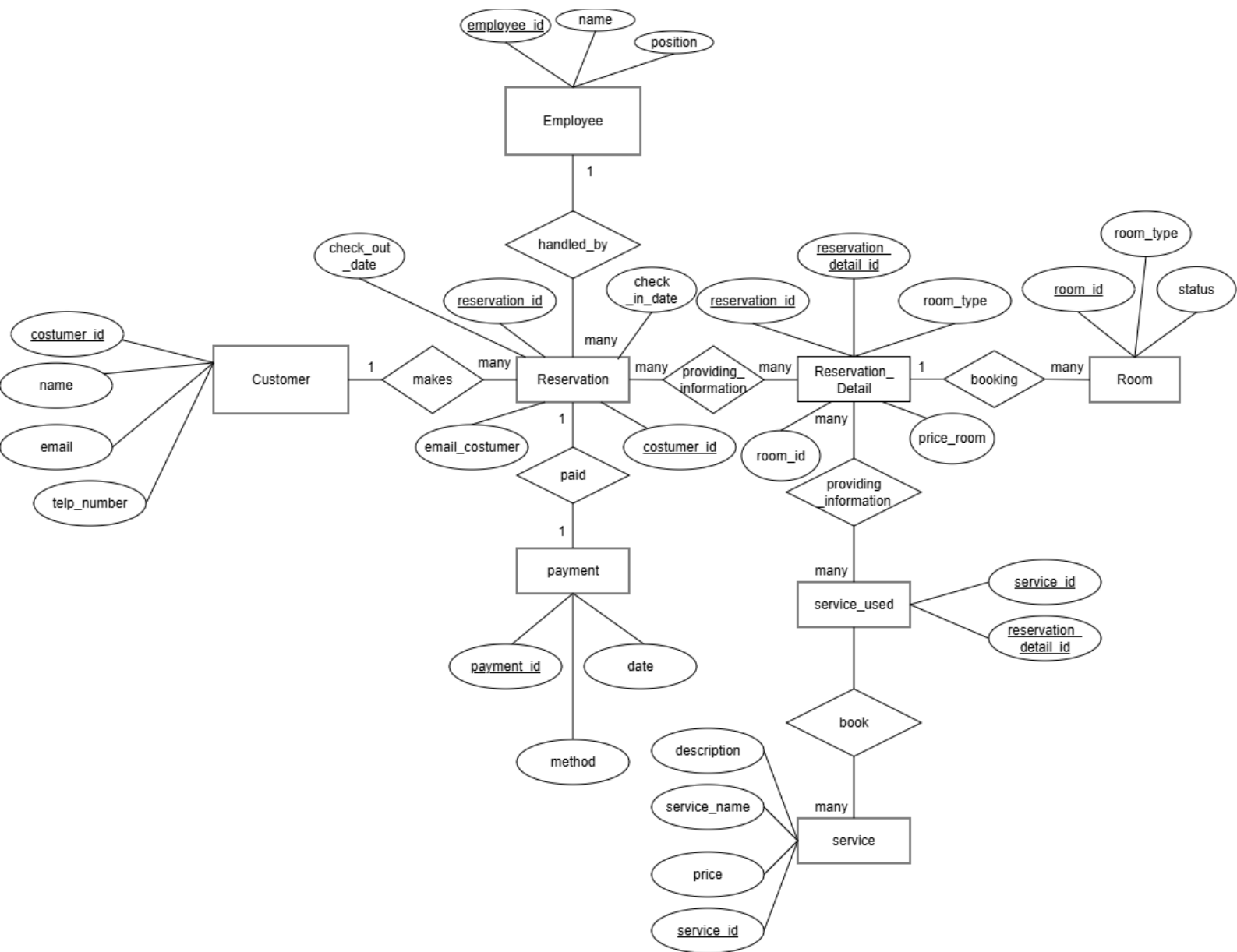
Customer id	Customer name	Email	Tele number	Reservation id	Reservation Detail id	Check In	Check Out	Employee id	Employee name	Position	Room id	Room Status	Room Type	Price Room	Service Name	Service Price	service id	description	Payment Id	Payment Method	Payment Date
G001	Andi	andi@gmail.com	8123456789	R001	RD01	01-09-25	02-09-25	E001	Sinta	Staff	RD01	Available	Deluxe	1000000	Food service	300000	S002	Dinner	PM001	Cash	01-09-25
G002	Budi	budi@mail.com	8987654321	R002	RD02	01-09-25	03-09-25	E001	Sinta	Staff	RD02	Occupied	Deluxe		Laundry service	50000	S004	2kg	PM002	Transfer	01-09-25
											ROs1	Available	Standart	750000	Laundry service	25000	S003	1kg			
G003	Citra	citra@gmail.com	8543285439	R003	RD03	02-09-25	04-09-25	E001	Sinta	Staff	ROs2	Available	Standart	750000	Laundry service	50000	S004	2kg	PM003	Transfer	02-09-25
											ROs3	Available	Standart		Food service	300000	S002	Dinner			
G004	Dika	dika@mail.com	8785499653	R004	RD04	02-09-25	05-09-25	E001	Sinta	Staff	RD03	Occupied	Deluxe	1000000	Room service	450000	S006	Clining service	PM004	Transfer	02-09-25
											RD04	Available	Deluxe		Laundry service	50000	S004	2kg			
											RD05	Occupied	Deluxe		Room service	100000	S005	extra bad			
											ROs2	Available	Suite	1500000	Room service	450000	S006	Clining service			
G005	Eka	eka@mail.com	8555555555	R005	RD05	03-09-25	04-05-25	E002	Reni	Assistant	ROs5	Occupied	Standart	750000	Food service	150000	S001	Breakfast	PM005	Cash	03-09-25
											ROs1	OCcupied	Standart		Laundry service	25000	S003	1kg			
											ROs3	Available	Suite	1500000	Room service	100000	S005	extra bad			
G006	Fajar	fajar@mail.com	8879543817	R006	RD06	05-09-25	06-09-25	E002	Reni	Assistant	ROs4	OCcupied	Suite		Laundry service	50000	S004	2kg	PM006	Cash	05-09-25
											ROs5	OCcupied	Suite		Food service	300000	S002	Dinner			
G007	Gina	gina@mail.com	8693506238	R007	RD07	07-09-25	09-09-2025	E003	Dita	Waiter	ROs2	Available	Standart	750000	Room service	450000	S006	Clining service	PM007	Cash	07-09-25
											ROs3	Available	Standart		Room service	100000	S005	extra bad			
G008	Gina	gina@mail.com	8776443663	R008	RD08	08-09-25	10-09-25	E004	Abrar	Staff	ROs1	Available	Suite	1500000	Room service	450000	S006	Clining service	PM008	Cash	08-09-25
											RD01	OCcupied	Deluxe	1000000	Laundry service	50000	S004	2kg			
G009	Ridho	ridho@mail.com	8676444325	R009	RD09	09-09	10-09-2025	E005	Fadil	Waiter	ROs4	Available	Standart	750000	Food service	150000	S001	Breakfast	PM009	Cash	09-09
											ROs5	Available	Standart		Laundry service	25000	S003	1kg			
											ROs1	Available	Standart		Room service	100000	S005	extra bad			
											ROs2	Available	Standart		Room service	450000	S006	Clining service			
G010	Aulia	aulia@mail.com	8977778643	R010	RD010	11-09-25	13-09-25	E001	Sinta	Staff	RD02	OCcupied	Deluxe	1000000	Food service	150000	S001	Breakfast	PM010	Cash	11-09-25
											RD03	Available	Deluxe		Food service	300000	S002	Dinner			
G011	Rizka	rizka@mail.com	8787878766	R011	RD011	14-09-25	15-09-25	E004	Abrar	Staf	ROs2	Available	Suite	1500000	Laundry service	25000	S003	1kg	PM011	Transfer	14-09-25
											ROs3	Available	Suite		Laundry service	50000	S004	2kg			
											ROs4	Available	Suite		Food service	300000	S002	Dinner			
											ROs5	OCcupied	Suite		Laundry service	25000	S003	1kg			
											ROs1	Available	Suite		Room service	100000	S005	extra bad			
											RD04	OCcupied	Deluxe	1000000	Laundry service	50000	S004	2kg			
G012	Muthia	muthia@mail.com	896754231	R012	RD012	15-09-25	16-09-25	E002	Reni	Assistant	RD05	OCcupied	Deluxe		Food service	300000	S002	Dinner	PM012	Transfer	15-09-25
											ROs2	Available	Suite	1500000	Room service	450000	S006	Clining service			
G013	Farah	farah@mail.com	897689765	R013	RD013	18-09-25	20-09-25	E003	Dita	Waiter	ROs2	Available	Suite	1500000	Room service	450000	S006	Clining service	PM013	Transfer	18-09-25
											RD01	Available	Deluxe	1000000	Laundry service	25000	S003	1kg			
G014	Aria	aria@mail.com	897665734	R014	RD014	21-09-25	22-09-25	E006	Ririn	Staff	RD02	Available	Deluxe		Food service	150000	S001	Breakfast	PM014	Cash	21-09-25
											RD03	OCcupied	Deluxe		Laundry service	50000	S004	2kg			
G015	Enran	enran@mail.com	898989	R015	RD015	25-09-25	26-09-25	E006	Ririn	Waiter	ROs3	OCcupied	Standart	750000	Room service	100000	S005	extra bad	PM015	Cash	25-09-25
											ROs3	OCcupied	Suite	1500000	Food service	150000	S006	Breakfast			

Customer_id	Customer_name	Email	Telp_number	Reservation_id	Reservation Detail ID	Check_In	Check_Out	Employee_id	Employee_name	Position	Room Id	Room Status	Room Type	Price Room	Service Name	Service Price	service id	description	Payment Id	Payment Method	Payment Date
G001	Andi	andi@mail.com	8123456789	R001	RD01	01-09-25	02-09-25	E001	Sinta	Staff	ROD1	Available	Deluxe	1000000	Food service	300000	S002	Dinner	PM001	Cash	01-09-25
G001	Andi	andi@mail.com	8123456789	R001	RD01	01-09-25	02-09-25	E001	Sinta	Staff	ROD2	Occupied	Deluxe	1000000	Laundry service	50000	S004	2kg	PM001	Cash	01-09-25
G002	Budi	budi@mail.com	8987654321	R002	RD02	01-09-25	03-09-25	E001	Sinta	Staff	ROs1	Available	Standart	750000	Laundry service	25000	S003	1kg	PM002	Transfer	01-09-25
G002	Budi	budi@mail.com	8987654321	R002	RD02	01-09-25	03-09-25	E001	Sinta	Staff	ROs1	Available	Suite	1500000	Food service	150000	S001	Breakfast	PM002	Transfer	01-09-25
G003	Citra	ctra@mail.com	8543285439	R003	RD03	02-09-25	04-09-25	E001	Sinta	Staff	ROs2	Available	Standart	750000	Laundry service	50000	S004	2kg	PM003	Transfer	02-09-25
G003	Citra	ctra@mail.com	8543285439	R003	RD03	02-09-25	04-09-25	E001	Sinta	Staff	ROs3	Available	Standart	750000	Food service	300000	S002	Dinner	PM003	Transfer	02-09-25
G003	Citra	ctra@mail.com	8543285439	R003	RD03	02-09-25	04-09-25	E001	Sinta	Staff	ROs4	Occupied	Standart	750000	Laundry service	25000	S003	1kg	PM003	Transfer	02-09-25
G004	Dika	dika@mail.com	8785499653	R004	RD04	02-09-25	05-09-25	E001	Sinta	Staff	ROD3	Occupied	Deluxe	1000000	Room service	450000	S005	Cleaning service	PM004	Transfer	02-09-25
G004	Dika	dika@mail.com	8785499653	R004	RD04	02-09-25	05-09-25	E001	Sinta	Staff	ROD4	Available	Deluxe	1000000	Laundry service	50000	S004	2kg	PM004	Transfer	02-09-25
G004	Dika	dika@mail.com	8785499653	R004	RD04	02-09-25	05-09-25	E001	Sinta	Staff	ROD5	Occupied	Deluxe	1000000	Room service	100000	S005	extra bed	PM004	Transfer	02-09-25
G004	Dika	dika@mail.com	8785499653	R004	RD04	02-09-25	05-09-25	E001	Sinta	Staff	ROS2	Available	Suite	1500000	Room service	450000	S005	Cleaning service	PM004	Transfer	02-09-25
G005	Eka	eka@mail.com	8555555555	R005	RD05	03-09-25	04-05-25	E002	Reni	Assistant	ROs5	Occupied	Standart	750000	Food service	150000	S001	Breakfast	PM005	Cash	03-09-25
G005	Eka	eka@mail.com	8555555555	R005	RD05	03-09-25	04-05-25	E002	Reni	Assistant	ROs1	Occupied	Standart	750000	Laundry service	25000	S003	1kg	PM005	Cash	03-09-25
G005	Eka	eka@mail.com	8555555555	R005	RD05	03-09-25	04-05-25	E002	Reni	Assistant	ROS3	Available	Suite	1500000	Room service	100000	S005	extra bad	PM005	Cash	03-09-25
G006	Fajar	fajar@mail.com	8879543817	R006	RD06	05-09-25	06-09-25	E002	Reni	Assistant	ROS4	Occupied	Suite	1500000	Laundry service	50000	S004	2kg	PM006	Cash	05-09-25
G006	Fajar	fajar@mail.com	8879543817	R006	RD06	05-09-25	06-09-25	E002	Reni	Assistant	ROS5	Occupied	Suite	1500000	Food service	300000	S002	Dinner	PM006	Cash	05-09-25
G007	Gina	gina@mail.com	8693506238	R007	RD07	07-09-25	09-09-2025	E003	Dita	Waiter	ROs2	Available	Standart	750000	Room service	450000	S006	Clining service	PM007	Cash	07-09-25
G007	Gina	gina@mail.com	8693506238	R007	RD07	07-09-25	09-09-2025	E003	Dita	Waiter	ROs3	Available	Standart	750000	Room service	100000	S005	extra bad	PM007	Cash	07-09-25
G008	Gina	gina@mail.com	8776443663	R008	RD08	08-09-25	10-09-25	E004	Abrar	Staff	ROs1	Available	Suite	1500000	Room service	450000	S006	Clining service	PM008	Cash	08-09-25
G008	Gina	gina@mail.com	8776443663	R008	RD08	08-09-25	10-09-25	E004	Abrar	Staff	ROD1	Occupied	Deluxe	1000000	Laundry service	50000	S004	2kg	PM008	Cash	08-09-25
G008	Gina	gina@mail.com	8776443663	R008	RD08	08-09-25	10-09-25	E004	Abrar	Staff	ROs4	Available	Standart	750000	Food service	150000	S001	Breakfast	PM008	Cash	08-09-25
G009	Ridho	ridho@mail.com	8676344325	R009	RD09	09-09	10-09-2025	E005	Fadil	Waiter	ROs5	Available	Standart	750000	Laundry service	25000	S003	1kg	PM009	Cash	09-09
G009	Ridho	ridho@mail.com	8676344325	R009	RD09	09-09	10-09-2025	E005	Fadil	Waiter	ROs1	Available	Standart	750000	Room service	100000	S005	extra bad	PM009	Cash	09-09
G009	Ridho	ridho@mail.com	8676344325	R009	RD09	09-09	10-09-2025	E005	Fadil	Waiter	ROs2	Available	Standart	750000	Room service	450000	S006	Clining service	PM009	Cash	09-09
G010	Aulia	aulia@mail.com	8977778643	R010	RD10	11-09-25	13-09-25	E001	Sinta	Staff	ROD2	Occupied	Deluxe	1000000	Food service	150000	S001	Breakfast	PM010	Cash	11-09-25
G010	Aulia	aulia@mail.com	8977778643	R010	RD10	11-09-25	13-09-25	E001	Sinta	Staff	ROD3	Available	Deluxe	1000000	Food service	300000	S002	Dinner	PM010	Cash	11-09-25
G011	Rizka	rizka@mail.com	8787878766	R011	RD11	14-09-25	15-09-25	E004	Abrar	Staff	ROS2	Available	Suite	1500000	Laundry service	25000	S003	1kg	PM011	Transfer	14-09-25
G011	Rizka	rizka@mail.com	8787878766	R011	RD11	14-09-25	15-09-25	E004	Abrar	Staff	ROS3	Available	Suite	1500000	Laundry service	50000	S004	2kg	PM011	Transfer	14-09-25
G011	Rizka	rizka@mail.com	8787878766	R011	RD11	14-09-25	15-09-25	E004	Abrar	Staff	ROS4	Available	Suite	1500000	Food service	300000	S002	Dinner	PM011	Transfer	14-09-25
G011	Rizka	rizka@mail.com	8787878766	R011	RD11	14-09-25	15-09-25	E004	Abrar	Staff	ROS5	Occupied	Suite	1500000	Laundry service	25000	S003	1kg	PM011	Transfer	14-09-25
G011	Rizka	rizka@mail.com	8787878766	R011	RD11	14-09-25	15-09-25	E004	Abrar	Staff	ROS1	Available	Suite	1500000	Room service	100000	S005	extra bad	PM011	Transfer	14-09-25
G012	Muthia	muthia@mail.com	896754231	R012	RD12	15-09-25	16-09-25	E002	Reni	Assistant	ROD4	Occupied	Deluxe	1000000	Laundry service	50000	S004	2kg	PM012	Transfer	15-09-25
G012	Muthia	muthia@mail.com	896754231	R012	RD12	15-09-25	16-09-25	E002	Reni	Assistant	ROD5	Occupied	Deluxe	1000000	Food service	300000	S002	Dinner	PM012	Transfer	15-09-25
G013	Farah	farah@mail.com	897689765	R013	RD13	18-09-25	20-09-25	E003	Dita	Waiter	ROS2	Available	Suite	1500000	Room service	450000	S006	Clining service	PM013	Transfer	18-09-25
G013	Farah	farah@mail.com	897689765	R013	RD13	18-09-25	20-09-25	E003	Dita	Waiter	ROD1	Available	Deluxe	1000000	Laundry service	25000	S003	1kg	PM013	Transfer	18-09-25
G014	Arla	arla@mail.com	897665734	R014	RD14	21-09-25	22-09-25	E006	Ririn	Staff	ROD2	Available	Deluxe	1000000	Food service	150000	S001	Breakfast	PM014	Cash	21-09-25
G014	Arla	arla@mail.com	897665734	R014	RD14	21-09-25	22-09-25	E006	Ririn	Staff	ROD3	Occupied	Deluxe	1000000	Laundry service	50000	S004	2kg	PM014	Cash	21-09-25
G015	Emran	emran@mail.com	898989	R015	RD15	25-09-25	26-09-25	E007	Riris	Waiter	ROs3	Occupied	Standart	750000	Room service	100000	S005	extra bad	PM015	Cash	25-09-25
G015	Emran	emran@mail.com	898989	R015	RD15	25-09-25	26-09-25	E007	Riris	Waiter	ROS3	Occupied	Suite	1500000	Food service	150000	S001	Breakfast	PM015	Cash	25-09-25

TABLE SERVICE				TABLE RESERVATION					TABLE ROOM				
service id	Service Name	Service Price	description		Reservation_id	Check_In	Check_Out	Room Id		Room Id	Room Status	Room Type	Price Room
S002	Food service	300000	Dinner		R001	01-09-25	02-09-25	ROD1		ROD1	Available	Deluxe	1000000
S004	Laundry service	50000	2kg		R001	01-09-25	02-09-25	ROD2		ROD2	Occupied	Deluxe	1000000
S003	Laundry service	25000	1kg		R002	01-09-25	03-09-25	ROs1		ROs1	Available	Standart	750000
S001	Food service	150000	Breakfast		R002	01-09-25	03-09-25	ROs1		ROs1	Available	Suite	1500000
S004	Laundry service	50000	2kg		R003	02-09-25	04-09-25	ROs2		ROs2	Available	Standart	750000
S002	Food service	300000	Dinner		R003	02-09-25	04-09-25	ROs3		ROs3	Available	Standart	750000
S003	Laundry service	25000	1kg		R003	02-09-25	04-09-25	ROs4		ROs4	Occupied	Standart	750000
S005	Room service	450000	Cleaning service		R004	02-09-25	05-09-25	ROD3		ROD3	Occupied	Deluxe	1000000
S004	Laundry service	50000	2kg		R004	02-09-25	05-09-25	ROD4		ROD4	Available	Deluxe	1000000
S005	Room service	100000	extra bed		R004	02-09-25	05-09-25	ROD5		ROD5	Occupied	Deluxe	1000000
S005	Room service	450000	Cleaning service		R004	02-09-25	05-09-25	ROS2		ROS2	Available	Suite	1500000
S001	Food service	150000	Breakfast		R005	03-09-25	04-05-25	ROs5		ROs5	Occupied	Standart	750000
S003	Laundry service	25000	1kg		R005	03-09-25	04-05-25	ROs1		ROs1	Occupied	Standart	750000
S005	Room service	100000	extra bad		R005	03-09-25	04-05-25	ROS3		ROS3	Available	Suite	1500000
S004	Laundry service	50000	2kg		R006	05-09-25	06-09-25	ROS4		ROS4	Occupied	Suite	1500000
S002	Food service	300000	Dinner		R006	05-09-25	06-09-25	ROS5		ROS5	Occupied	Suite	1500000
S006	Room service	450000	Clining service		R007	07-09-25	09-09-2025	ROs2		ROs2	Available	Standart	750000
S005	Room service	100000	extra bad		R007	07-09-25	09-09-2025	ROs3		ROs3	Available	Standart	750000
S006	Room service	450000	Clining service		R008	08-09-25	10-09-25	ROS1		ROS1	Available	Suite	1500000
S004	Laundry service	50000	2kg		R008	08-09-25	10-09-25	ROD1		ROD1	Occupied	Deluxe	1000000
S001	Food service	150000	Breakfast		R008	08-09-25	10-09-25	ROs4		ROs4	Available	Standart	750000
S003	Laundry service	25000	1kg		R009	09-09	10-09-2025	ROs5		ROs5	Available	Standart	750000
S005	Room service	100000	extra bad		R009	09-09	10-09-2025	ROs1		ROs1	Available	Standart	750000
S006	Room service	450000	Clining service		R009	09-09	10-09-2025	ROs2		ROs2	Available	Standart	750000
S001	Food service	150000	Breakfast		R010	11-09-25	13-09-25	ROD2		ROD2	Occupied	Deluxe	1000000
S002	Food service	300000	Dinner		R010	11-09-25	13-09-25	ROD3		ROD3	Available	Deluxe	1000000
S003	Laundry service	25000	1kg		R011	14-09-25	15-09-25	ROS2		ROS2	Available	Suite	1500000
S004	Laundry service	50000	2kg		R011	14-09-25	15-09-25	ROS3		ROS3	Available	Suite	1500000
S002	Food service	300000	Dinner		R011	14-09-25	15-09-25	ROS4		ROS4	Available	Suite	1500000
S003	Laundry service	25000	1kg		R011	14-09-25	15-09-25	ROS5		ROS5	Occupied	Suite	1500000
S005	Room service	100000	extra bad		R011	14-09-25	15-09-25	ROS1		ROS1	Available	Suite	1500000
S004	Laundry service	50000	2kg		R012	15-09-25	16-09-25	ROD4		ROD4	Occupied	Deluxe	1000000
S002	Food service	300000	Dinner		R012	15-09-25	16-09-25	ROD5		ROD5	Occupied	Deluxe	1000000
S006	Room service	450000	Clining service		R013	18-09-25	20-09-25	ROS2		ROS2	Available	Suite	1500000
S003	Laundry service	25000	1kg		R013	18-09-25	20-09-25	ROD1		ROD1	Available	Deluxe	1000000
S001	Food service	150000	Breakfast		R014	21-09-25	22-09-25	ROD2		ROD2	Available	Deluxe	1000000
S004	Laundry service	50000	2kg		R014	21-09-25	22-09-25	ROD3		ROD3	Occupied	Deluxe	1000000
S005	Room service	100000	extra bad		R015	25-09-25	26-09-25	ROs3		ROs3	Occupied	Standart	750000
S001	Food service	150000	Breakfast		R015	25-09-25	26-09-25	ROS3		ROS3	Occupied	Suite	1500000
	Reservation_id	Check_In	Check_Out	Room Id	service id								
	R001	01-09-25	02-09-25	ROD1	S002								
	R001	01-09-25	02-09-25	ROD2	S004								
	R002	01-09-25	03-09-25	ROs1	S003								
	R002	01-09-25	03-09-25	ROS1	S001								
	R003	02-09-25	04-09-25	ROs2	S004								
	R003	02-09-25	04-09-25	ROs3	S002								
	R003	02-09-25	04-09-25	ROs4	S003								

	R004	02-09-25	05-09-25	ROD3	S005								
	R004	02-09-25	05-09-25	ROD4	S004								
	R004	02-09-25	05-09-25	ROD5	S005								
	R004	02-09-25	05-09-25	ROS2	S005								
	R005	03-09-25	04-05-25	ROs5	S001								
	R005	03-09-25	04-05-25	ROs1	S003								
	R005	03-09-25	04-05-25	ROS3	S005								
	R006	05-09-25	06-09-25	ROS4	S004								
	R006	05-09-25	06-09-25	ROS5	S002								
	R007	07-09-25	09-09-2025	ROs2	S006								
	R007	07-09-25	09-09-2025	ROs3	S005								
	R008	08-09-25	10-09-25	ROS1	S006								
	R008	08-09-25	10-09-25	ROD1	S004								
	R008	08-09-25	10-09-25	ROs4	S001								
	R009	09-09	10-09-2025	ROs5	S003								
	R009	09-09	10-09-2025	ROs1	S005								
	R009	09-09	10-09-2025	ROs2	S006								
	R010	11-09-25	13-09-25	ROD2	S001								
	R010	11-09-25	13-09-25	ROD3	S002								
	R011	14-09-25	15-09-25	ROS2	S003								
	R011	14-09-25	15-09-25	ROS3	S004								
	R011	14-09-25	15-09-25	ROS4	S002								
	R011	14-09-25	15-09-25	ROS5	S003								
	R011	14-09-25	15-09-25	ROS1	S005								
	R012	15-09-25	16-09-25	ROD4	S004								
	R012	15-09-25	16-09-25	ROD5	S002								
	R013	18-09-25	20-09-25	ROS2	S006								
	R013	18-09-25	20-09-25	ROD1	S003								
	R014	21-09-25	22-09-25	ROD2	S001								
	R014	21-09-25	22-09-25	ROD3	S004								
	R015	25-09-25	26-09-25	ROs3	S005								
	R015	25-09-25	26-09-25	ROS3	S001								

RESERVATION DETAIL				ROOM				CUSTOMER				EMPLOYEE				SERVICE				RESERVATION				PAYMENT				
Reservation ID	Room ID	reservation detail	Email Customer	room type	price room	Room ID	Room Type	Room Status	Customer ID	Customer Name	Temp Number	Email	Employee ID	Employee Name	Position	Service ID	Service name	Description	Price	Reservation	Check In	Check Out	Customer ID	Email Customer	Payment ID	Payment Method	Payment Date	
R001	R001	R001	ada@gmail.com	Deluxe	1500000	R0A1	Standard	Available	G001	Jack	8123456789	jack@gmail.com	E001	Steve	Staff	S001	Food Service	Breakfast	1500000	R001	01-09-25	04-09-25	G001	ada@gmail.com	P0001	Cash	01-09-25	
R002	R001	R001	bad@gmail.com	Standard	750000	R0A2	Standard	Available	G002	Bob	8887654321	bad@gmail.com	E002	Ravi	Assistant	S002	Dinner	Dinner	300000	R002	01-09-25	03-09-25	G002	bad@gmail.com	P0002	Transfer	01-09-25	
R003	R003	R003	citra@gmail.com	Standard	750000	R0A3	Standard	Available	G003	Citra	8543265439	citra@gmail.com	E003	Dita	Waiter	S003	Luxury Service	Dinner	450000	R003	02-09-25	04-09-25	G003	citra@gmail.com	P0003	Transfer	02-09-25	
R004	R003	R004	dasa@gmail.com	Deluxe	1500000	R0A4	Standard	Occupied	G004	Daja	8785499653	dasa@gmail.com	E004	Akbar	Staff	S004	2 Inj		80000	R004	02-09-25	04-09-25	G004	dasa@gmail.com	P0004	Transfer	02-09-25	
R005	R005	R005	esaa@gmail.com	Standard	750000	R0A5	Standard	Available	G005	Ela	8355505555	esaa@gmail.com	E005	Fadh	Waiter	S005	Room Service	Luxury Bed	100000	R005	03-09-25	04-05-25	G005	esaa@gmail.com	P0005	Cash	03-09-25	
R006	R006	R006	fajar@gmail.com	Suite	1000000	R0D1	Deluxe	Available	G006	Eja	8879543817	fajar@gmail.com	E006	Rini	Staff	S006	Cleaning Service		4500000	R006	05-09-25	06-09-25	G006	fajar@gmail.com	P0006	Cash	05-09-25	
R007	R007	R007	gaa@gmail.com	Standard	750000	R0D2	Deluxe	Occupied	G007	Gina	8993506238	gaa@gmail.com	E007	Rini	Waiter					R007	07-09-25	09-09-2025	G007		P0007	Cash	07-09-25	
R008	R008	R008	hadi@gmail.com	Suite	1000000	R0D3	Deluxe	Occupied	G008	Hani	8776433663	hadi@gmail.com				S008				R008	08-09-25	10-09-25	G008	hadi@gmail.com	P0008	Cash	08-09-25	
R009	R009	R009	info@gmail.com	Standard	750000	R0D4	Deluxe	Available	G009	Ridho	8876544325	info@gmail.com				S009				R009	09-09	10-09-2025	G009	info@gmail.com	P0009	Cash	11-09-25	
R010	R002	R010	ada@gmail.com	Deluxe	1500000	R0D5	Deluxe	Available	G010	Adha	8977789543	ada@gmail.com				S010				R010	11-09-25	13-09-25	G010	ada@gmail.com	P0010	Cash	11-09-25	
R011	R011	R011	ada@gmail.com	Deluxe	1500000	R0S1	Suite	Available	G011	Adha	8781878187	ada@gmail.com				S011				R011	14-09-25	15-09-25	G011	ada@gmail.com	P0011	Transfer	14-09-25	
R012	R004	R012	muha@gmail.com	Deluxe	1000000	R0S2	Suite	Available	G012	Muha	896754231	muha@gmail.com				S012				R012	15-09-25	16-09-25	G012	muha@gmail.com	P0012	Transfer	15-09-25	
R013	R002	R013	fajar@gmail.com	Suite	1000000	R0S3	Suite	Available	G013	Farah	8988989765	fajar@gmail.com				S013				R013	18-09-25	20-09-25	G013	fajar@gmail.com	P0013	Transfer	18-09-25	
R014	R014	R014	esaa@gmail.com	Deluxe	1500000	R0S4	Suite	Occupied	G014	Esa	8978657890	esaa@gmail.com				S014				R014	21-09-25	23-09-25	G014	esaa@gmail.com	P0014	Cash	21-09-25	
R015	R015	R015	emran@gmail.com	Standard	750000	R0S5	Suite	Occupied	G015	Emran	898898	emran@gmail.com				S015				R015	25-09-25	26-09-25	G015	emran@gmail.com	P0015	Cash	25-09-25	
SERVICE USED																												
reservation detail id	service id																											
R001	S002																											
R002	S003																											
R003	S004																											
R004	S006																											
R005	S001																											
R006	S004																											
R007	S006																											
R008	S006																											
R009	S003																											
R010	S001																											
R011	S003																											
R012	S004																											
R013	S006																											
R014	S001																											
R015	S003																											





UNIVERSITAS
INDONESIA

CEP-CCIT
FAKULTAS TEKNIK

**RELATIONAL DATABASE DESIGN FOR SYSTEM MANAGEMENT
HOTEL**

Arranged by:

Muhammad Ridho Abrar	2521010189
Arla Bukit Elbirun	2521010177
Farisa Aulia Sudrajat	2521010184

CEP CCIT-FTUI
FACULTY OF ENGINEERING
UNIVERSITY OF INDONESIA
2025

PROJECT ON
RELATIONAL DATABASE DESIGN FOR SYSTEM MANAGEMENT
HOTEL

Developed by

- 1. Muhammad Ridho Abrar**
- 2. Arla Bukit Elbirun**
- 3. Farisa Aulia Sudrajat**

RELATIONAL DATABASE DESIGN FOR SYSTEM MANAGEMENT HOTEL

Batch code: 1CC7

Start code: 14 December 2025

End code: 24 December 2025

Name of Faculty: Muhammad Riza Iqbal

Names of Developer:

1. Muhammad Ridho Abrar
2. Arla Bukit Elbirun
3. Farisa Aulia Sudrajat

Date of Submission: 24 December 2025

CERTIFICATE

This report titled “ Relational Database Design for Hotel Management System” embodies the original work done by Arla Bukit Elbirun, Farisa Aulia Sudrajat, and Muhammad Ridho Abrar. Project in partial fulfillment of their course requirement at CEP-CCIT FTUI.

Coordinator:
Muhammad Riza Iqbal

ACKNOWLEDGEMENT

First of all, thank you to Allah SWT because with Allah's help, the author has successfully completed writing a project entitled “Relational Database Design for Hotel Management Systems” right on time. The purpose of writing this paper is to fulfill the assignment given by Mr. Hary Prasetyo Rosianto, S.Tr.T. and Mr. Riza Iqbal Latief, S.T. as lecturers in the Information Technology Study Program.

In compiling this project, the author faced many challenges and obstacles, but with the help of many people, these obstacles were overcome. The author also realizes that there are still many mistakes in the process of working on this project. Therefore, the author would like to thank all the individuals who have helped in the process of working on this project.

May God reward all your help and bless you all. The author realizes that this project is still imperfect in terms of its structure and content. Therefore, the author hopes that criticism from readers can help the author in perfecting the next project. Last but not least, hopefully this project can help readers to gain further knowledge about the Information Technology study program.

BACKGROUND

The development of the hotel industry demands a management system capable of handling various operational activities, such as maintaining reservations, guest information, and payment processing. Using manual methods often creates obstacles, such as recording errors, repetitive data, and inefficient decision making processes.

Implementing a relational database offers a solution because it can store information in a structured manner in the form of interconnected tables. Through this system, various entities such as guests, rooms, reservations, and payment transactions can be managed in an integrated manner. Furthermore, processes such as real-time room availability checks and revenue analysis can be performed more easily. A properly designed relational database will streamline hotel operations, accelerate decision making, and improve shared service. Therefore, designing a comprehensive, efficient, and scalable database is essential to support all hotel activities.

SYSTEM ANALYSIS

The analysis of the relational database design for the hotel management system emphasises the importance of managing operational hotel data such as guest, room, reservation and payment data in an integrated manner. Manual systems or unstructured databases can lead to data duplication, recording errors, and slower operational processes.

Relational database management systems organise data into tables that are linked using primary keys and foreign keys. The main entities are designed according to the hotel's business processes, thus supporting data normalisation, maintaining consistency, and facilitating processes such as checking room availability and managing transactions.

With a well-designed relational database, hotel systems become more efficient, secure, and easy to develop. This helps management to make informed decisions and improve the quality of service to guests.

DATABASE DESIGN

Database Name : HotelReservation

Number of tables: 8

- Costumer Table
- Employee Table
- Room Table
- Service Table
- Reservation Table
- Reservation detail table
- Service Used Table
- Payment Table

Number of View: 1

- vw_reservation_summary

Number of Procedure: 3

- proc_AddReservation
- proc_AddServiceUsed
- proc_getReservationDetailWithServices

Number of Triggers: 4

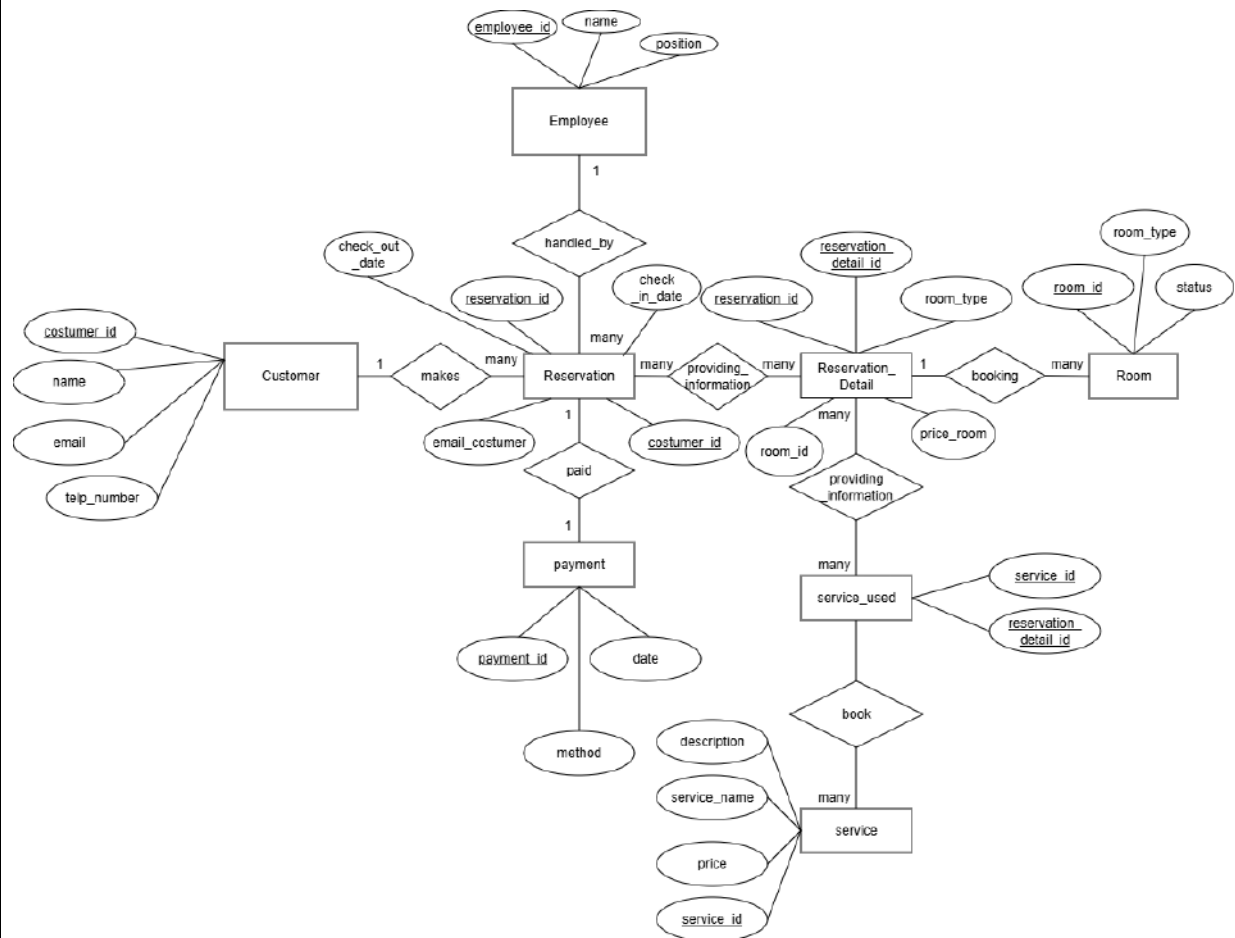
- Trg_after_insert_reservation
- Trg_after_delete_reservation
- trg_after_insert_service_used
- trg_after_delete_service_used

- trg_before_insert_service_used
- Trg_before_insert_reservation
- Trg_after_delete_reservation

Number of Index: 5

- Idx_name
- ft_idx_customer_name

ERD DIAGRAM



NORMALIZATION TABLES

In ERD there is normalization where each element depends on the primary key to connect them between each table. The following pictures are Normalization Tables from our Management Hotel.

3NF

Customer			
Customer_ID	Customer_Name	Telp_Number	Email
G001	Andi	8123456789	andi@mail.com
G002	Budi	8987654321	budi@mail.com
G003	Citra	8543285439	citra@mail.com

RESERVATION				
Reservation_ID	Check_In	Check_Out	Customer_ID	Email_Customer
R001	01-09-25	02-09-25	G001	andi@mail.com
R002	01-09-25	03-09-25	G002	budi@mail.com
R003	02-09-25	04-09-25	G003	citra@mail.com

RESERVATION_DETAIL					
Reservation_ID	room_id	reservation_detail_id	Email_Customer	room_type	price_room
R001	ROD1	RD01	andi@mail.com	Deluxe	1500000
R002	ROs1	RD02	budi@mail.com	Standart	750000
R003	ROs2	RD03	citra@mail.com	Standart	750000

ROOM		
Room_ID	Room_Type	Room_Status
ROs1	Standart	Available
ROs2	Standart	Available
ROs3	Standart	Available

Employee		
Employee_ID	Employee_Name	Position
E001	Sinta	Staff
E002	Reni	Assistant
E003	Dita	Walter

SERVICE_USED	
reservation_detail_id	service_id
RD01	S002
RD02	S003
RD03	S004

Service			
Service_ID	Service_name	Description	Price
S001	Food Service	Breakfast	150000
S002		Dinner	300000
S003	Loundry Service	1 kg	25000
S004		2 kg	50000

PAYMENT		
Payment_ID	Payment_Method	Payment_Date
PM001	Cash	01-09
PM002	Transfer	01-09-25
PM003	Transfer	02-09-25

ERD DESIGN

In the database there is a top down approach, namely ERD tables, which is to implement the database. The following pictures are ERD Tables from our Management Hotel.

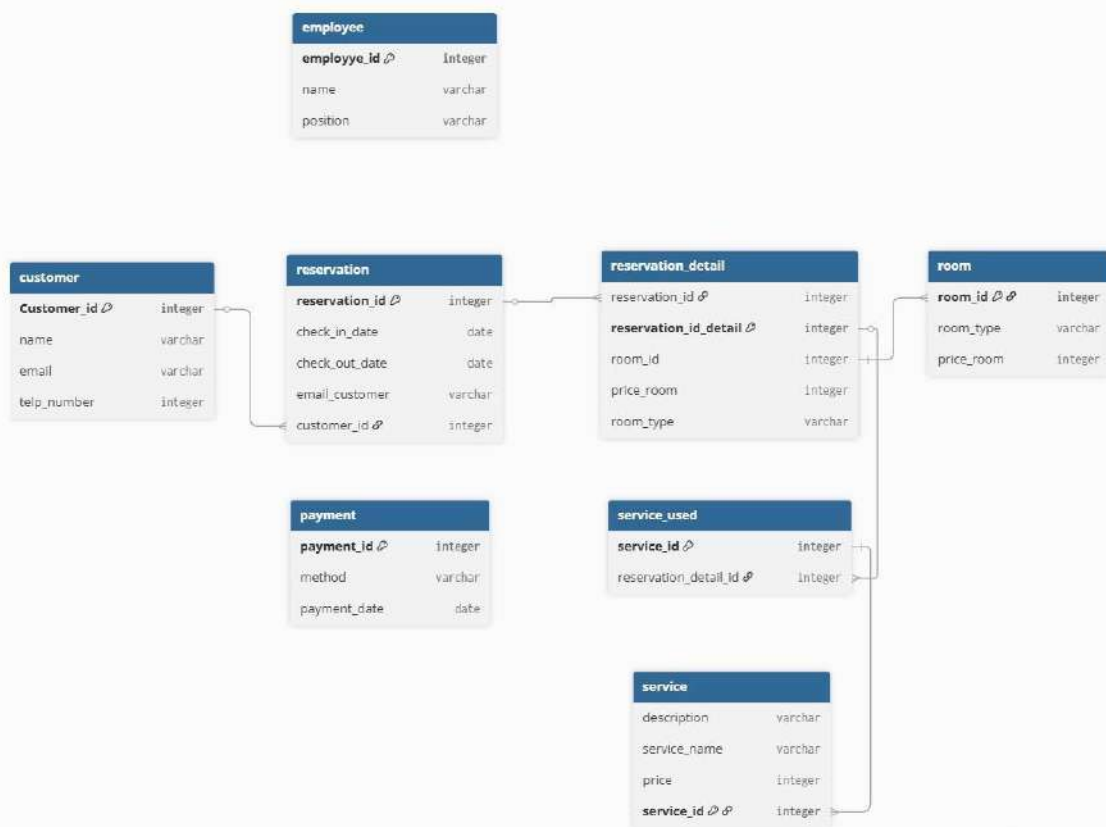


TABLE DESIGN

The table design of the Management Hotel database is as follows:

Table Costumer

	Customer_id	Customer_name	Email	Telp_number
▶	G001	Andi	andi@mail.com	8123456789
	G002	Budi	budi@mail.com	8987654321
	G003	Citra	citra@mail.com	8543285439
	G004	Dika	dika@mail.com	8785499653
	G005	Eka	eka@mail.com	8555555555
★	NULL	NULL	NULL	NULL

Table Reservation

	Reservation_id	Check_in	Check_out	Customer_id	Email_Customer	Room_id
▶	R001	2025-09-01	2025-09-02	G001	andi@mail.com	ROD1
	R002	2025-09-01	2025-09-03	G002	budi@mail.com	ROS1
	R003	2025-09-03	2025-09-04	G001	andi@mail.com	ROs2
	R004	2025-09-05	2025-09-06	G002	budi@mail.com	ROD3
	R005	2025-09-07	2025-09-08	G001	andi@mail.com	ROs5
	R006	2025-09-08	2025-09-09	G002	andi@gmail.com	ROD1

Table Reservation Detail

	Reservation_detail_id	Reservation_id	Room_id	Room_type	Price_room
▶	RD001	R001	ROD1	Deluxe	1000000.00
	RD002	R002	ROS1	Standard	750000.00
	RD003	R003	ROs2	Standard	750000.00
	RD004	R004	ROD3	Deluxe	1000000.00
	RD005	R005	ROs5	Standard	750000.00
★	NULL	NULL	NULL	NULL	NULL

Table Room

	Room_id	Room_status	Room_type
▶	ROD1	Occupied	Deluxe
	ROD3	Available	Deluxe
	ROS1	Available	Standard
	ROs2	Available	Standard
	ROs5	Available	Standard
★	NULL	NULL	NULL

Table Employee

	Employee_id	Employee_name	Position
▶	E001	Sinta	Staff
	E002	Reni	Assistant
	E003	Dita	Waiter
	E005	Fadil	Waiter
★	NULL	NULL	NULL

TABLE DESIGN

Table Service Used

	Reservation_detail_id	Service_id
▶	RD001	S001
	RD002	S002
	RD003	S004
	RD004	S005
	RD005	S001
•	NULL	NULL

Table Service

	Service_id	Service_name	Service_price	Description
▶	S001	Food service	300000.00	Dinner
	S002	Laundry service	50000.00	2kg
	S003	Laundry_service	25000.00	1kg
	S004	Food_Service	150000.00	Breakfast
	S005	Room_Service	100000.00	Extra_bed
•	NULL	NULL	NULL	NULL

Table Room

	Payment_id	Payment_method	Payment_date
▶	PM001	Cash	2025-09-01
	PM002	Transfer	2025-09-01
	PM003	Transfer	2025-09-02
	PM004	Transfer	2025-09-02
	PM005	Cash	2025-09-03
•	NULL	NULL	NULL

VALIDATION PERFORM

Costumer Table

Column	Method(s) for user validation	Description
Costumer_id	PRIMARY KEY, NOT NULL	This column contains primary key, this column cannot be empty
Costumer_name	NOT NULL	This Column cannot be empty
Email	NOT NULL	This Column cannot be empty
Telp_Number	NOT NULL	This Column cannot be empty

Employee Table

Column	Method(s) for user validation	Description
Employee_id	PRIMARY KEY, NOT NULL	This column contains primary key, this column cannot be empty
Employee_Name	NOT NULL	This Column cannot be empty
Position	NOT NULL	This Column cannot be empty

Room Table

Column	Method(s) for user validation	Description
Room_id	PRIMARY KEY, NOT NULL	This column contains primary key, this column cannot be empty
Room_status	NOT NULL	This Column cannot be empty
Room_type	NOT NULL	This Column cannot be empty

TABLE DESIGN

Service Table

Column	Method(s) for user validation	Description
Service_id	PRIMARY KEY, NOT NULL	This column contains primary key, this column cannot be empty
Service_name	NOT NULL	This Column cannot be empty
Service_price	NOT NULL	This Column cannot be empty

Reservation Table

Column	Method(s) for user validation	Description
Reservation_id	PRIMARY KEY, NOT NULL	This column contains primary key, this column cannot be empty
Check_in	NOT NULL	This Column cannot be empty
Check_out	NOT NULL	This Column cannot be empty
Costumer_id	FOREIGN KEY, NOT NULL	This column contains foreign key references from costumer, this column cannot be empty
Email_Costumer	NOT NULL	This Column cannot be empty
Room_id	NOT NULL	This Column cannot be empty

TABLE DESIGN

Reservation detail table

Column	Method(s) for user validation	Description
Reservation_detail_id	PRIMARY KEY, NOT NULL	This column contains primary key, this column cannot be empty
Reservation_id	FOREIGN KEY, NOT NULL	This column contains foreign key references from reservation, this column cannot be empty
Room_id	FOREIGN KEY, NOT NULL	This column contains foreign key references from room, this column cannot be empty
Room_type	NOT NULL	This column cannot be empty
Price_Room	NOT NULL	This column cannot be empty

Service used table

Column	Method(s) for user validation	Description
Reservation_detail_id	PRIMARY KEY, FOREIGN KEY, NOT NULL	This column contains primary key, This column contains foreign key references from Reservation_detail, This column cannot be empty
Service_id	FOREIGN KEY, NOT NULL	This column contains foreign key references from service, This Column cannot be empty

Payment Table

Column	Method(s) for user validation	Description
Payment_id	PRIMARY KEY, NOT NULL	This column contains primary key, this column cannot be empty
Payment_method	NOT NULL	This Column cannot be empty
Payment_date	NOT NULL	This Column cannot be empty

STORE PROCEDURE

Query of Procedure Add Reservation:

```
DELIMITER $$
```

```
CREATE PROCEDURE make_reservation(  
    IN p_reservation_id VARCHAR(10),  
    IN p_check_in DATE,  
    IN p_check_out DATE,  
    IN p_customer_id VARCHAR(10),  
    IN p_email_customer VARCHAR(50),  
    IN p_room_id VARCHAR(10)  
)  
BEGIN  
    DECLARE v_conflict_count INT;  
  
    SELECT COUNT(*) INTO v_conflict_count  
    FROM reservation  
    WHERE Room_id = p_room_id  
        AND NOT (Check_out <= p_check_in OR Check_in >=  
p_check_out);  
  
    IF v_conflict_count > 0 THEN  
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Room  
already booked for the selected dates.';  
    ELSE  
        INSERT INTO reservation (Reservation_id,
```

```

    Check_in, Check_out, Customer_id, Email_Customer, Room_id)
        VALUES (p_reservation_id, p_check_in, p_check_out,
p_customer_id, p_email_customer, p_room_id);

        UPDATE room SET Room_status = 'Occupied' WHERE
Room_id = p_room_id;
    END IF;
END $$

DELIMITER ;

DELIMITER $$

```

Query of Procedure Add Service Used:

```

CREATE PROCEDURE addServiceUsed (
    IN p_reservation_detail_id VARCHAR(10),
    IN p_service_id VARCHAR(10)
)
BEGIN
    IF NOT EXISTS (SELECT 1 FROM reservation_detail WHERE
reservation_detail_id = p_reservation_detail_id) THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT =
'reservation_detail_id tidak ditemukan.';
    END IF;

    IF NOT EXISTS (SELECT 1 FROM service WHERE service_id =
p_service_id) THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'service_id
tidak ditemukan.';
    END IF;
END

```

```
END IF;
```

```
        INSERT INTO service_used (reservation_detail_id,  
service_id)  
        VALUES (p_reservation_detail_id, p_service_id);  
END $$
```

```
DELIMITER ;
```

```
DELIMITER $$
```

Query of procedure Reservation Detail With Services:

```
CREATE PROCEDURE getReservationDetailWithServices (  
    IN p_reservation_id VARCHAR(10)  
)  
BEGIN  
    SELECT rd.reservation_detail_id, rd.room_id, rd.room_type,  
rd.price_room,  
        s.service_id, s.service_name, s.service_price  
    FROM reservation_detail rd  
    LEFT JOIN service_used su ON rd.reservation_detail_id =  
su.reservation_detail_id  
    LEFT JOIN service s ON su.service_id = s.service_id  
    WHERE rd.reservation_id = p_reservation_id;  
END $$
```

```
DELIMITER ;
```

VIEW

```
1 • SELECT * FROM hotelreservation.vw_reservation_summary;
```

reservation_id	customer_name	check_in	check_out	room_type	service_name	service_price
R001	Andi	2025-09-01	2025-09-02	Deluxe	Food service	300000.00
R002	Budi	2025-09-01	2025-09-03	Standard	Laundry service	50000.00
R003	Andi	2025-09-03	2025-09-04	Standard	Food_Service	150000.00
R004	Budi	2025-09-05	2025-09-06	Deluxe	Room_Service	100000.00
R005	Andi	2025-09-07	2025-09-08	Standard	Food service	300000.00

Query:

```
1 • CREATE VIEW vw_reservation_summary AS
2   SELECT r.reservation_id, c.customer_name, r.check_in, r.check_out, rd.room_type, s.service_name, s.service_price
3   FROM reservation r
4   JOIN customer c ON r.customer_id = c.customer_id
5   JOIN reservation_detail rd ON r.reservation_id = rd.reservation_id
6   JOIN service_used su ON rd.reservation_detail_id = su.reservation_detail_id
7   JOIN service s ON su.service_id = s.service_id;
```

TRIGGER

TRIGGER UPDATE AFTER INSERT RESERVATION

```
DELIMITER $$
```

```
CREATE TRIGGER trg_after_insert_reservation
```

```
AFTER INSERT ON reservation
```

```
FOR EACH ROW
```

```
BEGIN
```

```
    UPDATE room SET room_status = 'Occupied' WHERE room_id =  
NEW.room_id;
```

```
END $$
```

```
DELIMITER ;
```

```
DELIMITER $$
```

TRIGGER UPDATE AFTER DELETE RESERVATION

```
CREATE TRIGGER trg_after_delete_reservation
```

```
AFTER DELETE ON reservation
```

```
FOR EACH ROW
```

```
BEGIN
```

```
    UPDATE room
```

```
    SET room_status = 'Available'
```

```
    WHERE room_id = OLD.Room_id;
```

END \$\$

DELIMITER ;

DELIMITER \$\$

TRIGGER UPDATE AFTER INSERT SERVICE USED

```
CREATE TRIGGER trg_after_insert_service_used
AFTER INSERT ON service_used
FOR EACH ROW
BEGIN
    UPDATE reservation_detail rd
    SET rd.total_service_price = (
        SELECT IFNULL(SUM(s.service_price), 0)
        FROM service_used su
        JOIN service s ON su.service_id = s.service_id
        WHERE su.reservation_detail_id =
NEW.reservation_detail_id
    )
    WHERE rd.reservation_detail_id = NEW.reservation_detail_id;
END $$
```

TRIGGER UPDATE AFTER DELETE SERVICE USED

```
CREATE TRIGGER trg_after_delete_service_used
AFTER DELETE ON service_used
FOR EACH ROW
BEGIN
    UPDATE reservation_detail rd
    SET rd.total_service_price = (
        SELECT IFNULL(SUM(s.service_price), 0)
```

```

        FROM service_used su
        JOIN service s ON su.service_id = s.service_id
        WHERE su.reservation_detail_id =
OLD.reservation_detail_id
    )
    WHERE rd.reservation_detail_id = OLD.reservation_detail_id;
END $$

```

```

DELIMITER ;

```

```

DELIMITER $$

```

TRIGGER UPDATE BEFORE INSERT SERVICE USED

```

CREATE TRIGGER trg_before_insert_service_used
BEFORE INSERT ON service_used
FOR EACH ROW
BEGIN
    DECLARE v_count INT;

    -- Cek reservation_detail_id ada
    SELECT COUNT(*) INTO v_count FROM reservation_detail WHERE
reservation_detail_id = NEW.reservation_detail_id;
    IF v_count = 0 THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Invalid
reservation_detail_id';
    END IF;

    -- Cek service_id ada
    SELECT COUNT(*) INTO v_count FROM service WHERE service_id
= NEW.service_id;

```

```
    IF v_count = 0 THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Invalid
service_id';
    END IF;
END $$
```

```
DELIMITER $$
```

TRIGGER UPDATE BEFORE INSERT RESERVATION

```
CREATE TRIGGER trg_before_insert_reservation
BEFORE INSERT ON reservation
FOR EACH ROW
BEGIN
    DECLARE v_conflict_count INT;

    SELECT COUNT(*) INTO v_conflict_count
    FROM reservation
    WHERE Room_id = NEW.Room_id
        AND NOT (Check_out <= NEW.Check_in OR Check_in >=
NEW.Check_out);

    IF v_conflict_count > 0 THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Room
already booked for the selected dates.';
    END IF;
END $$

DELIMITER ;

DELIMITER $$
```

TRIGGER VALIDATION AFTER DELETE RESERVATION

```
CREATE TRIGGER trg_after_delete_reservation
AFTER DELETE ON reservation
FOR EACH ROW
BEGIN
    DECLARE v_room_booked_count INT;

    SELECT COUNT(*) INTO v_room_booked_count
    FROM reservation
    WHERE Room_id = OLD.Room_id
    AND Check_out > CURDATE();

    IF v_room_booked_count = 0 THEN
        UPDATE room SET Room_status = 'Available' WHERE Room_id
= OLD.Room_id;
    END IF;
END $$

DELIMITER ;
```

FULL TEXT SEARCH

Full Text Search

```
use hotelreservation;
```

```
ALTER TABLE Customer  
ADD FULLTEXT INDEX ft_idx_customer_name  
(Customer_name);
```

```
SELECT * FROM Customer  
WHERE MATCH(Customer_name)  
AGAINST ('Andi' IN NATURAL LANGUAGE MODE);
```

Index

	Customer_id	Customer_name	Email	Telp_number
▶	G001	Andi	andi@mail.com	8123456789
•	NULL	NULL	NULL	NULL

CONFIGURATION

Hardware : Laptop

Operating System : Windows 11

Software : Microsoft Word 2019

MYSQL Workbench

	PROJECT FILE DETAILS	
No	File Name	Remarks
1	Project group 1.pdf	PDF that contains report
2	HotelReservation.sql	Database File
3	HOTEL PPT.pdf	PDF that contains presentation slide

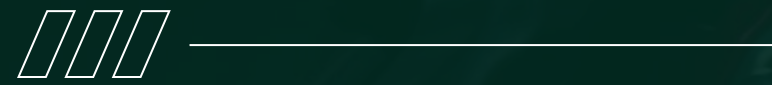
RELATIONAL DATABASE DESIGN FOR HOTEL MANAGEMENT SYSTEM

Presented By: Arla Bukit Elbirun

Farisa Aulia Sudrajat

Muhammad Ridho Akbar

ENGINEERING



BACKGROUND

The use of manual methods in hotel management often leads to recording errors and obstacles in decision-making. Relational databases are available to systematically integrate various entities such as reservations, guests, and payments. Through this system, hotel operations become easier due to instant revenue analysis and room availability monitoring. An efficient database design can support business growth and more professional services.



SYSTEM ANALYSIS

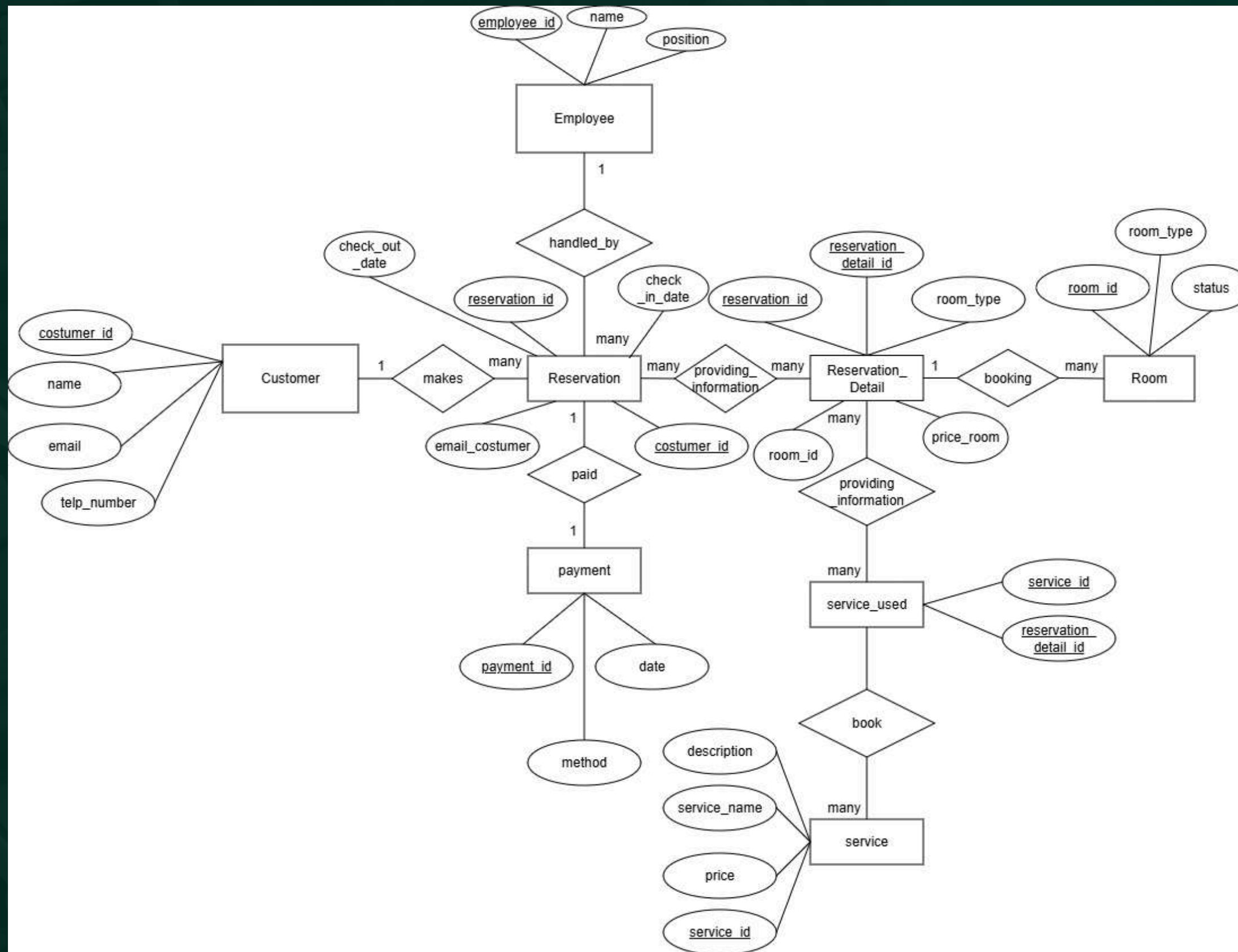


The relational database system integrates all hotel operational data into structured tables to overcome the weaknesses of manual systems, which are slow and prone to errors. The use of relational keys in entity design ensures that data remains consistent, accurate, and easy to manage in accordance with hotel business processes. As a result, operations become more efficient and secure, making it easier for management to monitor transactions and improve overall service standards for guests.





ERD DESIGN





NORMALIZATION TABLE





Next!

MYSQL DEMO





THANK YOU



+123-456-7890



@reallygreatsite



reallygreatsite.com

THANK YOU

■ CREATE DATABASE

```
CREATE DATABASE IF NOT EXISTS HotelReservation;
```

```
USE HotelReservation;
```

```
CREATE TABLE Customer (
```

```
    Customer_id VARCHAR(10) NOT NULL,
```

```
    Customer_name VARCHAR(100) NOT NULL,
```

```
    Email VARCHAR(100) NOT NULL,
```

```
    Telp_number VARCHAR(15) NOT NULL,
```

```
    PRIMARY KEY (Customer_id),
```

```
    UNIQUE (Email),
```

```
    UNIQUE (Telp_number)
```

```
);
```

```
CREATE TABLE Employee (
```

```
    Employee_id VARCHAR(10) NOT NULL,
```

```
    Employee_name VARCHAR(100) NOT NULL,
```

```
    Position VARCHAR(50) NOT NULL,
```

```
    PRIMARY KEY (Employee_id)
```

```
);
```

```
CREATE TABLE Room (
```

```
    Room_id VARCHAR(10) NOT NULL,
```

```
    Room_status VARCHAR(20) NOT NULL,
```

```
    Room_type VARCHAR(20) NOT NULL,
```

```
    PRIMARY KEY (Room_id),
```

```
    CHECK (Room_status IN ('Available', 'Occupied'))
```

```
);
```

```
CREATE TABLE Service (
```

```
    Service_id VARCHAR(10) NOT NULL,
```

```
    Service_name VARCHAR(100) NOT NULL,
```

```
    Service_price DECIMAL(10,2) NOT NULL,
```

```
    Description VARCHAR(255),
```

```

        PRIMARY KEY (Service_id)
    );

CREATE TABLE Reservation (
    Reservation_id VARCHAR(10) NOT NULL,
    Check_in DATE NOT NULL,
    Check_out DATE NOT NULL,
    Customer_id VARCHAR(10) NOT NULL,
    Email_Customer VARCHAR(50) NOT NULL,
    Room_id VARCHAR(10),
    PRIMARY KEY (Reservation_id),
    FOREIGN KEY (Customer_id) REFERENCES Customer(Customer_id),
    CHECK (Check_out > Check_in)
);

CREATE TABLE Reservation_Detail (
    Reservation_detail_id VARCHAR(10) NOT NULL,
    Reservation_id VARCHAR(10) NOT NULL,
    Room_id VARCHAR(10) NOT NULL,
    Room_type VARCHAR(20) NOT NULL,
    Price_room DECIMAL(10,2) NOT NULL,
    PRIMARY KEY (Reservation_detail_id),
    FOREIGN KEY (Reservation_id) REFERENCES Reservation(Reservation_id),
    FOREIGN KEY (Room_id) REFERENCES Room(Room_id)
);

CREATE TABLE Service_Used (
    Reservation_detail_id VARCHAR(10) NOT NULL,
    Service_id VARCHAR(10) NOT NULL,
    PRIMARY KEY (Reservation_detail_id, Service_id),
    FOREIGN KEY (Reservation_detail_id) REFERENCES
Reservation_Detail(Reservation_detail_id),
    FOREIGN KEY (Service_id) REFERENCES Service(Service_id)
);

```

```

CREATE TABLE Payment (
    Payment_id VARCHAR(10) NOT NULL,
    Payment_method VARCHAR(20) NOT NULL,
    Payment_date DATE NOT NULL,
    PRIMARY KEY (Payment_id),
    CHECK (Payment_method IN ('Cash', 'Transfer'))
);

CREATE INDEX IDX_Customer_Email ON Customer(Email);
CREATE INDEX IDX_Customer_Name ON Customer(Customer_name);
CREATE INDEX IDX_Reservation_Check_In ON Reservation(Check_in);
CREATE INDEX IDX_Reservation_Customer_ID ON Reservation(Customer_id);
CREATE INDEX IDX_Payment_Date ON Payment(Payment_date);

INSERT INTO Customer (Customer_id, Customer_name, Email, Telp_number)
VALUES
('G001', 'Andi', 'andi@mail.com', '8123456789'),
('G002', 'Budi', 'budi@mail.com', '8987654321');

INSERT INTO Employee (Employee_id, Employee_name, Position) VALUES
('E001', 'Sinta', 'Staff'),
('E002', 'Reni', 'Assistant');

INSERT INTO Room (Room_id, Room_status, Room_type) VALUES
('ROD1', 'Available', 'Deluxe'),
('ROS1', 'Available', 'Standard');

INSERT INTO Service (Service_id, Service_name, Service_price, Description)
VALUES
('S001', 'Food service', 300000.00, 'Dinner'),
('S002', 'Laundry service', 50000.00, '2kg');

INSERT INTO Reservation (Reservation_id, Check_in, Check_out, Customer_id,
Email_Customer, Room_id) VALUES
('R001', '2025-09-01', '2025-09-02', 'G001', 'andi@mail.com', 'ROD1'),
('R002', '2025-09-01', '2025-09-03', 'G002', 'budi@mail.com', 'ROS1');

```

```
INSERT INTO Reservation_Detail (Reservation_detail_id, Reservation_id,
Room_id, Room_type, Price_room) VALUES
```

```
('RD001', 'R001', 'ROD1', 'Deluxe', 1000000.00),
('RD002', 'R002', 'ROS1', 'Standard', 750000.00);
```

```
INSERT INTO Service_Used (Reservation_detail_id, Service_id) VALUES
```

```
('RD001', 'S001'),
('RD002', 'S002');
```

```
INSERT INTO Payment (Payment_id, Payment_method, Payment_date) VALUES
```

```
('PM001', 'Cash', '2025-09-01'),
('PM002', 'Transfer', '2025-09-01');
```

■ TRIGGER

```
DELIMITER $$
```

```
CREATE TRIGGER trg_after_insert_reservation
```

```
AFTER INSERT ON reservation
```

```
FOR EACH ROW
```

```
BEGIN
```

```
    UPDATE room SET room_status = 'Occupied' WHERE room_id = NEW.room_id;
```

```
END $$
```

```
DELIMITER ;
```

```
DELIMITER $$
```

```
CREATE TRIGGER trg_after_delete_reservation
```

```
AFTER DELETE ON reservation
```

```
FOR EACH ROW
```

```
BEGIN
```

```
    UPDATE room
```

```
    SET room_status = 'Available'
```

```
        WHERE room_id = OLD.Room_id;
END $$
```

```
DELIMITER ;
```

```
DELIMITER $$
```

```
CREATE TRIGGER trg_after_insert_service_used
AFTER INSERT ON service_used
FOR EACH ROW
BEGIN
    UPDATE reservation_detail rd
    SET rd.total_service_price = (
        SELECT IFNULL(SUM(s.service_price), 0)
        FROM service_used su
        JOIN service s ON su.service_id = s.service_id
        WHERE su.reservation_detail_id = NEW.reservation_detail_id
    )
    WHERE rd.reservation_detail_id = NEW.reservation_detail_id;
END $$
```

```
CREATE TRIGGER trg_after_delete_service_used
AFTER DELETE ON service_used
FOR EACH ROW
BEGIN
    UPDATE reservation_detail rd
    SET rd.total_service_price = (
        SELECT IFNULL(SUM(s.service_price), 0)
        FROM service_used su
        JOIN service s ON su.service_id = s.service_id
        WHERE su.reservation_detail_id = OLD.reservation_detail_id
    )
    WHERE rd.reservation_detail_id = OLD.reservation_detail_id;
END $$
```

DELIMITER ;

DELIMITER \$\$

CREATE TRIGGER trg_before_insert_service_used

BEFORE INSERT ON service_used

FOR EACH ROW

BEGIN

DECLARE v_count INT;

SELECT COUNT(*) INTO v_count FROM reservation_detail WHERE
reservation_detail_id = NEW.reservation_detail_id;

IF v_count = 0 THEN

SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Invalid
reservation_detail_id';

END IF;

SELECT COUNT(*) INTO v_count FROM service WHERE service_id =
NEW.service_id;

IF v_count = 0 THEN

SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Invalid service_id';

END IF;

END \$\$

DELIMITER \$\$

CREATE TRIGGER trg_before_insert_reservation

BEFORE INSERT ON reservation

FOR EACH ROW

BEGIN

DECLARE v_conflict_count INT;

SELECT COUNT(*) INTO v_conflict_count

FROM reservation

WHERE Room_id = NEW.Room_id

```

        AND NOT (Check_out <= NEW.Check_in OR Check_in >= NEW.Check_out);

    IF v_conflict_count > 0 THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Room already booked for
the selected dates.';
    END IF;
END $$

DELIMITER ;

DELIMITER $$

CREATE TRIGGER trg_after_delete_reservation
AFTER DELETE ON reservation
FOR EACH ROW
BEGIN
    DECLARE v_room_booked_count INT;

    SELECT COUNT(*) INTO v_room_booked_count
    FROM reservation
    WHERE Room_id = OLD.Room_id
    AND Check_out > CURDATE();

    IF v_room_booked_count = 0 THEN
        UPDATE room SET Room_status = 'Available' WHERE Room_id =
OLD.Room_id;
    END IF;
END $$

DELIMITER ;

DELIMITER $$

```

■ PROCEDURE

```
CREATE PROCEDURE make_reservation(  
    IN p_reservation_id VARCHAR(10),  
    IN p_check_in DATE,  
    IN p_check_out DATE,  
    IN p_customer_id VARCHAR(10),  
    IN p_email_customer VARCHAR(50),  
    IN p_room_id VARCHAR(10)  
)  
  
BEGIN  
    DECLARE v_conflict_count INT;  
  
    SELECT COUNT(*) INTO v_conflict_count  
    FROM reservation  
    WHERE Room_id = p_room_id  
    AND NOT (Check_out <= p_check_in OR Check_in >= p_check_out);  
  
    IF v_conflict_count > 0 THEN  
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Room already booked for  
the selected dates.';  
    ELSE  
        INSERT INTO reservation (Reservation_id, Check_in,  
Check_out, Customer_id, Email_Customer, Room_id)  
        VALUES (p_reservation_id, p_check_in, p_check_out, p_customer_id,  
p_email_customer, p_room_id);  
  
        UPDATE room SET Room_status = 'Occupied' WHERE Room_id =  
p_room_id;  
    END IF;  
END $$  
  
DELIMITER ;  
  
DELIMITER $$  
  
CREATE PROCEDURE addServiceUsed (  

```

```

        IN p_reservation_detail_id VARCHAR(10),
        IN p_service_id VARCHAR(10)
    )
BEGIN
    IF NOT EXISTS (SELECT 1 FROM reservation_detail WHERE
reservation_detail_id = p_reservation_detail_id) THEN

        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'reservation_detail_id
tidak ditemukan.';

        END IF;

    IF NOT EXISTS (SELECT 1 FROM service WHERE service_id = p_service_id)
THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'service_id tidak
ditemukan.';

        END IF;

    INSERT INTO service_used (reservation_detail_id, service_id)
    VALUES (p_reservation_detail_id, p_service_id);
END $$

DELIMITER ;

DELIMITER $$

CREATE PROCEDURE getReservationDetailWithServices (
    IN p_reservation_id VARCHAR(10)
)
BEGIN
    SELECT rd.reservation_detail_id, rd.room_id, rd.room_type,
rd.price_room,
        s.service_id, s.service_name, s.service_price
    FROM reservation_detail rd
    LEFT JOIN service_used su ON rd.reservation_detail_id =
su.reservation_detail_id
    LEFT JOIN service s ON su.service_id = s.service_id
    WHERE rd.reservation_id = p_reservation_id;
END $$

```

DELIMITER ;

■ VIEW

```
CREATE VIEW vw_reservation_summary AS

SELECT r.reservation_id, c.customer_name, r.check_in, r.check_out,
rd.room_type, s.service_name, s.service_price

FROM reservation r

JOIN customer c ON r.customer_id = c.customer_id

JOIN reservation_detail rd ON r.reservation_id = rd.reservation_id

JOIN service_used su ON rd.reservation_detail_id = su.reservation_detail_id

JOIN service s ON su.service_id = s.service_id;
```

```
CREATE INDEX idx_reservation_detail_reservation_id
ON reservation_detail(reservation_id);
```

```
CREATE INDEX idx_service_service_id
ON service(service_id);
```

```
CREATE INDEX idx_service_service_name
ON service(service_name);
```

```
CREATE INDEX idx_service_used_reservation_detail_service
ON service_used(reservation_detail_id, service_id);
```

```
CREATE INDEX idx_reservation_customer_id
ON reservation(customer_id);
```

■ FULLTEXT SEARCH

```
use hotelreservation;
```

```
ALTER TABLE Customer ADD FULLTEXT INDEX ft_idx_customer_name
(Customer_name);
```

```
SELECT * FROM Customer WHERE MATCH(Customer_name) AGAINST ('Andi' IN
NATURAL LANGUAGE MODE);
```